

Risk-Aware Relational Spectral Recovery under Partial Quantum Observability

Keywords: relational spectral recovery; partial observability; quantum error correction; recovery control; quantum information; risk-aware control

Abstract

Quantum recovery under partial observability must operate with incomplete, noisy, or ambiguous diagnostic evidence. In this setting, recovery policies may improve average logical fidelity while still introducing rare but severe lower-tail failures through unsafe candidate switches. We introduce a finite and auditable framework for risk-aware relational spectral recovery that targets this failure mode. The method constructs internally clock-conditioned system slices and maps each slice to a weighted relational graph. The ordered family of graph Laplacian spectra defines a relational spectral representation used for admissibility-aware recovery. A syndrome-first candidate and a relational candidate are then evaluated by a policy layer. The robust controller C_3^R treats the balanced policy C_2 as a proposal and applies a one-sided veto under excessive admissibility violation or syndrome uncertainty. The framework is evaluated on encoded three-qubit repetition-code simulations under clean, partial, noisy, partial-plus-noisy, and ambiguity-driven syndrome regimes. The main benefit appears in the lower tail rather than in a large shift of the mean. In degraded regimes, C_3^R reduces false-safe failures and improves lower-tail recovery behavior while remaining inactive in clean and noisy-only negative controls. These results support a restricted operating-zone interpretation: the proposed framework acts as a conservative recovery-control layer rather than as a decoder replacement or a foundations-level theory of quantum time.

1 Introduction

Quantum recovery increasingly has to operate under incomplete and imperfect diagnostic information. In encoded quantum systems, syndrome measurements provide the primary evidence for selecting a correction, but this evidence may be missing, corrupted, ambiguous, or affected by measurement and reset errors. Standard recovery and mitigation frameworks address important aspects of this problem through recovery maps, syndrome-based correction, autonomous protection, probabilistic cancellation, and learning-based error mitigation [1, 2, 3, 4, 5]. However, when observability is partial, mean fidelity improvement is an insufficient criterion. A recovery policy may improve the average logical fidelity while still producing rare but severe lower-tail failures: a proposed correction that the degraded syndrome evidence appeared to support turns out to worsen the protected state. These unsafe candidate switches are the central failure mode this work targets. An intervention mechanism that can selectively identify and veto unsafe proposed switches—without disrupting reliable recovery under adequate syndrome evidence—would provide a conservative safety layer above syndrome-first recovery pipelines.

We propose to construct such an intervention signal from internally clocked conditional system slices. At each simulation step, we condition a global quantum state on a discrete internal clock register and extract a system state for each clock label. Each conditional slice is mapped to a weighted relational graph, and the ordered family of graph Laplacian spectra across clock labels defines a relational spectral trajectory: a compact, auditable, row-by-row structural summary of the encoded system indexed by internal labels rather than an external time variable. A proposed recovery candidate can be tested for structural consistency within this trajectory space—a check we call admissibility—before the policy layer decides whether to accept or veto a proposed switch. This construction is operationally motivated by relational clock formalisms, in particular Page–Wootters-type conditioning [6, 7, 8], but we do not use it to establish a foundations-level theory of quantum time or relational dynamics. The clock register is a minimal finite object, and the resulting trajectory is used strictly as a recovery-side diagnostic.

To make this structural evidence operational, we represent each clock-conditioned slice as a weighted relational graph and summarize it through its graph Laplacian spectrum. Spectral graph methods provide a natural way to capture relational connectivity, perturbation structure, and collective behaviour, and prior work has connected graph Laplacian spectra with quantum-state

representations and quantum network structure [9, 10, 11]. The key object in our framework is not a single graph spectrum but a clock-indexed spectral family: an ordered collection of Laplacian spectra that can be compared against a reference bank, audited for admissibility, and used to test whether a proposed recovery remains structurally consistent across the full set of internal clock labels. A local spectral anomaly that creates discontinuities across neighboring clock slices is a stronger diagnostic signal than any single-snapshot syndrome check.

The present study addresses a gap between structured recovery-side evidence and the decision layer that uses it. Characterization and benchmarking methods quantify device behavior and validate performance claims [12], and uncertainty-quantification frameworks emphasize calibrated interpretation of finite or incomplete observations [13, 14]. These tools operate primarily on syndrome records, channel models, circuit-level characterizations, or externally indexed data. Quantum recovery, mitigation, and noise-adaptation methods provide further tools for noisy processors, but they also typically treat syndrome information as the primary and sufficient evidence source [15, 16, 17]. The present work connects these directions by formulating recovery as admissible restoration in a clock-conditioned relational spectral trajectory space, and evaluating whether the resulting structural signal can serve as an auditable intervention mechanism when syndrome evidence is degraded.

We evaluate this framework in a controlled encoded-QEC testbed. Each simulation row produces two recovery candidates: a syndrome-first decoded recovery (candidate *A*) and a relational admissible recovery (candidate *B*). A policy layer then decides whether a proposed switch to the relational candidate should be accepted or vetoed. The revised controller C_3^R is constructed as a one-sided veto on the balanced proposal policy C_2 : it can block an unsafe proposed switch to *B*, but cannot create additional relational switches. This design makes the intervention auditable and the tradeoff explicit—one can ask whether the controller remains inactive in reliable regimes, whether it selectively blocks harmful proposed switches under degraded syndrome evidence, and whether the prevented harmful-switch loss exceeds the missed beneficial gain.

The main contribution is therefore a finite, auditable, risk-aware recovery-control framework built on relational spectral evidence under partial quantum observability. We define clock-conditioned spectral trajectories, introduce admissibility constraints and a reference-anchored recovery objective, and evaluate the one-sided veto controller under clean, noisy-only, partial, partial-plus-noisy, and ambiguity-plus-measurement syndrome stress protocols. The results support a restricted operating-zone interpretation: C_3^R remains inactive in clean and noisy-only controls, but under partial-plus-noisy observability reduces the false-safe fidelity rate from 6.85% to 1.34%, shifts the lower 5% gain quantile to zero, and improves lower-tail conditional value at risk. Intervention diagnostics show that prevented harmful-switch loss exceeds missed beneficial gain in the active regimes. The framework therefore provides a conservative operational baseline demonstrating that relational spectral admissibility can serve as a recovery-side safety signal under partial quantum observability, without claiming to replace optimized decoders or to establish a foundations-level theory of quantum time. We present this work as the first in a planned sequence: subsequent studies will examine whether the same admissibility and risk-control geometry generalizes to nontrivial stabilizer codes, and whether the relational spectral trajectory space may support learned structural representations of admissible recovery geometry.

2 Background and Positioning

2.1 Clock-conditioned relational dynamics

Relational formulations of quantum dynamics describe change through correlations with an internal clock rather than an externally supplied time parameter. This idea appears in background-independence and relational-dynamics discussions [18, 19]. The Page–Wootters mechanism gives the operational ingredient used here: a constrained global state can be conditioned on a clock subsystem to obtain relative system states [6, 7, 8]. Related work has studied robustness, interactions, noncausal circuits, measurement-related time flow, and computational uses of quantum time [20, 21, 22, 23, 24]. Our use of this literature is intentionally narrow. We do not attempt to solve the problem of time or derive continuum dynamics from a finite clock; we use clock-conditioned slices as finite relational data objects for recovery control.

2.2 Spectral graph representations of quantum structure

Graph-based and spectral representations provide compact descriptions of relational structure. In quantum information, graph Laplacians have been related to density matrices, separability, and quantum-state representations [9, 10, 25]. More broadly, quantum network models provide a

language for connecting graph structure, correlations, and quantum dynamics [11]. These works motivate the use of spectral signatures as auditable summaries of relational connectivity.

The present work differs from graph-state or network modeling approaches in its use of clock-conditioned spectral families. A single graph spectrum is not the central object. Instead, each internal clock label produces a relational graph, and the ordered family of Laplacian spectra defines a relational spectral trajectory. This trajectory is then compared with reference-compatible families and audited through admissibility penalties before it enters the recovery policy layer.

2.3 Quantum recovery, mitigation, and control

Quantum recovery and error mitigation have developed along several complementary directions. Recovery-map approaches formalize channel-level reconstruction and approximate reversibility [1], while quantum error-correction experiments and autonomous protection schemes demonstrate how encoded or stabilized systems can preserve quantum information under noise [26, 2, 27]. In the NISQ setting, mitigation methods such as probabilistic error cancellation, virtual distillation, detection-based mitigation, and learning-based correction address noisy circuit outputs without requiring full fault tolerance [3, 15, 16, 4, 5, 17].

The proposed framework is not a replacement for these methods. It is closer to a recovery-control layer placed above candidate recovery mechanisms. In the encoded-QEC experiments, the syndrome-first recovery remains the decoder-facing baseline, while the relational admissible recovery supplies an alternative candidate. The policy layer asks when a switch to that candidate should be trusted, when it should be vetoed, and how the tradeoff should be evaluated under partial observability.

2.4 Partial observability, uncertainty, and benchmarking

Reliable recovery decisions depend on how much diagnostic information is available and how trustworthy that information is. Benchmarking and characterization methods provide tools for quantifying device behavior and validating performance claims [12]. Randomized-measurement techniques and uncertainty-quantification frameworks also emphasize that finite, noisy, or incomplete observations must be interpreted with calibrated uncertainty [13, 14]. Noise modeling, live noise fingerprinting, robust control, and non-Markovian noise studies further show that quantum behavior can depend strongly on the structure and timing of noise processes [28, 29, 30, 31, 32, 33].

The present study focuses on a narrower but related issue: unsafe candidate selection when syndrome evidence is partial or jointly corrupted. Rather than claiming a calibrated posterior uncertainty model, we use a deliberately simple syndrome-stress index and evaluate its behavior through threshold and uncertainty-bin diagnostics. This makes the controller auditable and exposes the tradeoff between preventing harmful switches and mistakenly blocking beneficial ones.

3 Methods

This section develops the proposed framework in five steps. We first define the clock-conditioned relational setting. We then construct relational spectral signatures from each clock slice and assemble them into a spectral trajectory. Next, we formulate recovery as admissible restoration toward a reference-compatible spectral family. Finally, we instantiate the framework in an encoded-QEC testbed and define the risk-aware metrics used for evaluation. The methodological separation is important. The relational formulation specifies where the data objects come from; the spectral representation specifies what is measured; the recovery objective specifies how a corrupted relational trajectory is restored; and the encoded-QEC protocol tests whether the restored trajectory should be trusted when direct syndrome evidence is incomplete. This prevents the controller from being interpreted as either a new universal decoder or a claim that all clock-conditioned dynamics have been solved. The narrower claim is that relational spectral evidence can be made auditable and can be combined with syndrome evidence through an explicit downside-risk policy. The five-step separation also defines the interfaces along which subsequent work can substitute components without altering the surrounding decision geometry. Figure 1 summarizes the full pipeline, with the four numbered stages keyed to the subsections that follow.

3.1 Clock-conditioned relational formulation

We use a clock-conditioned relational formulation inspired by background-independent quantum dynamics and relational-time constructions [34, 6, 8]. The total Hilbert space is decomposed as

$$\mathcal{H}_{\text{tot}} = \mathcal{H}_C \otimes \mathcal{H}_S, \quad (1)$$

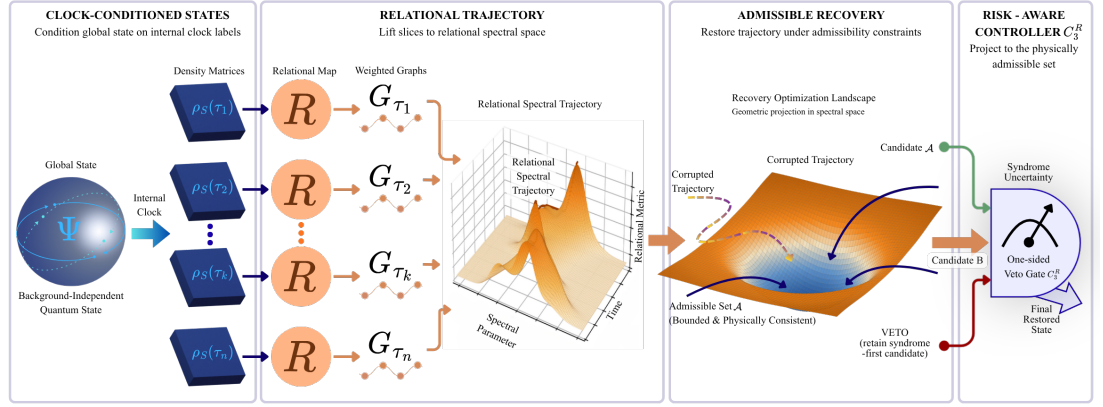


Figure 1. Pipeline overview of the proposed framework. A constrained global state is conditioned on internal clock labels to obtain system slices $\rho_S(\tau_r)$. Each slice is mapped to a weighted relational graph and reduced to a Laplacian spectral signature, yielding the relational spectral trajectory Γ_Ψ . The observed trajectory is restored through a reference-anchored admissible recovery objective, producing the relational candidate B . At the policy layer, C_3^R acts as a one-sided veto that can retain the syndrome-first candidate A or accept the C_2 -proposed switch to B .

where $\mathcal{H}_C$ is an internal clock register and $\mathcal{H}_S$ is the system register. Rather than assuming an externally supplied time coordinate, we work with a constrained global state

$$\hat{H}_{\text{tot}}|\Psi\rangle = 0. \quad (2)$$

We use this global constraint not as a claim that the finite simulation realizes a complete Wheeler–DeWitt theory or quantum-cosmological dynamics [35, 36, 37], but as a construction principle for generating internally indexed conditional system slices. In the computational pipeline, the role of the constraint is to define the admissible clock-conditioned family $\{\rho_S(\tau_r)\}_{r=1}^m$, from which all relational spectra, recovery objectives, and policy diagnostics are derived. In a finite-dimensional simulation, the existence and resolution of clock labels depend on spectral matching and recurrence constraints; those conditions are treated as consistency requirements and are moved to the Supplementary Information.

For a clock label τ , the clock-conditioned system state is

$$\rho_S(\tau) = \frac{\text{Tr}_C[(|\tau\rangle\langle\tau|_C \otimes I_S) |\Psi\rangle\langle\Psi|]}{\text{Tr}[(|\tau\rangle\langle\tau|_C \otimes I_S) |\Psi\rangle\langle\Psi|]}. \quad (3)$$

We therefore do not model recovery as a trajectory over an externally supplied background time coordinate. Instead, we condition a timeless global state on internal clock labels and construct recovery evidence from the resulting relational slices. Detailed consistency conditions for the global constraint, finite-dimensional clock representation, and clock-resolution assumptions are provided in Supplementary Note S1. In the implementation, each encoded simulation instance uses one clock qubit with initial state $|+\rangle_C$ and clock Hamiltonian $H_C = 0.8X_C$. The clock grid is uniform with

$$m = 9, \quad \tau_r \in \{-2.0, -1.5, \dots, 1.5, 2.0\}. \quad (4)$$

The conditional states $\rho_S(\tau_r)$ are materialized from the simulated global density matrix using Eq. (3). Unless otherwise stated, the same clock grid, clock Hamiltonian, and conditional-slice construction are used for all policies and stress protocols. This finite sampling is also why the framework is naturally compatible with partial observability: a degraded experiment may hide or corrupt part of the evidence available to the policy layer without changing the underlying definition of a clock-conditioned slice. The one-qubit clock is used deliberately as a minimal finite relational clock, not as a continuum-clock approximation [38, 20]. This choice keeps the validation focused on the minimal controlled question: whether clock-conditioned relational spectra can provide auditable recovery evidence before scaling to larger clock registers or denser relational clocks.

3.2 Relational spectral representation

For each clock-conditioned state $\rho_S(\tau)$, we build a weighted relational network whose edge weights encode pairwise relational evidence between system degrees of freedom,

$$\begin{aligned} W_{ij}^{(\tau)} &= f(\rho_S(\tau); i, j), \\ W_{ij}^{(\tau)} &= W_{ji}^{(\tau)}, \quad W_{ij}^{(\tau)} \geq 0. \end{aligned} \quad (5)$$

The reported experiments use explicit pairwise functionals rather than leaving f unspecified. Let $\rho_{ij}^{(\tau)}$ be the two-qubit reduction of $\rho_S(\tau)$ and let $\rho_i^{(\tau)}$ be the one-qubit reduction. For phase-flip-code experiments we use the absolute two-qubit coherence score

$$W_{ij}^{(\tau)} = \sum_{a \neq b} |(\rho_{ij}^{(\tau)})_{ab}|, \quad i \neq j, \quad (6)$$

in the computational basis. For bit-flip-code experiments we use the von Neumann mutual-information score

$$W_{ij}^{(\tau)} = S(\rho_i^{(\tau)}) + S(\rho_j^{(\tau)}) - S(\rho_{ij}^{(\tau)}), \quad (7)$$

with negative numerical round-off clipped to zero. Alternative pairwise functionals are possible and are treated as representation-robustness questions for future work, but all reported policy comparisons use the fixed code-dependent mapping above. The two choices reflect the dominant relational signature of the corresponding encoded basis. Phase-flip protection is sensitive to phase coherence, so the computational-basis coherence functional directly measures off-diagonal relational structure. Bit-flip protection is associated with population-basis error structure, for which mutual information provides a basis-independent pairwise dependence score. Because this choice can affect sensitivity, we treat representation robustness as a limitation rather than claiming uniqueness of the graph functional. We use a graph Laplacian rather than raw edge weights because its spectrum summarizes collective connectivity, is stable under node relabeling, and allows perturbations to be compared through standard spectral quantities [39, 9, 10].

The corresponding degree matrix and graph Laplacian are

$$D_{ii}^{(\tau)} = \sum_{j \neq i} W_{ij}^{(\tau)}, \quad L(\tau) = D^{(\tau)} - W^{(\tau)}. \quad (8)$$

Let the ordered Laplacian spectrum be

$$\begin{aligned} \Lambda(\tau) &= (\lambda_1(\tau), \dots, \lambda_n(\tau)), \\ \lambda_1(\tau) &\leq \dots \leq \lambda_n(\tau). \end{aligned} \quad (9)$$

The central object is not an individual graph or a single spectrum, but the clock-conditioned spectral family

$$\Gamma_\Psi : T \rightarrow \mathbb{R}^n, \quad \Gamma_\Psi(\tau) = \Lambda(\tau). \quad (10)$$

We use the term trajectory to denote an ordered family over internal clock labels, not evolution with respect to external background time. This distinction matters for the recovery task. A local perturbation at one clock slice may be harmless if it is compatible with the neighboring slices, but the same perturbation can be suspicious if it creates a discontinuity in the spectral family. The object being recovered is therefore the whole clock-indexed spectral signature, not a collection of independent spectra. This representation is intended to expose relational inconsistencies that may not be captured by a single syndrome snapshot.

Computationally, the collection of admissible spectral families is treated as a finite metric trajectory representation. For two spectral families $\Gamma^{(a)}$ and $\Gamma^{(b)}$ sampled at clock labels $\{\tau_r\}_{r=1}^m$, we use

$$\begin{aligned} d_{\mathcal{M}}^{ab} &\equiv d_{\mathcal{M}}(\Gamma^{(a)}, \Gamma^{(b)}), \\ d_{\mathcal{M}}^{ab} &= \left(\sum_{r=1}^m w_r \left\| \Lambda^{(a)}(\tau_r) - \Lambda^{(b)}(\tau_r) \right\|_2^2 \right)^{1/2}. \end{aligned} \quad (11)$$

The geometric terminology used below should be read as a formal interpretation of this finite metric trajectory representation. The detailed tangent-space, geodesic, and perturbation arguments are given in Supplementary Notes S2 and S3. The weights w_r allow the implementation to emphasize or de-emphasize particular clock slices, but the experiments reported here use the same metric form across all policies. Thus differences between C_1 , C_2 , C_3 , and C_3^R arise from the decision layer, not from changing the underlying trajectory distance.

3.3 Admissible spectral recovery

Let Γ_{obs} be the observed spectral trajectory after noise or partial observation. Recovery is posed as a constrained restoration problem in spectral-trajectory space. We define an admissible set

$$\mathcal{A} = \mathcal{A}_{\text{lap}} \cap \mathcal{A}_{\text{rel}} \cap \mathcal{A}_{\text{ref}}, \quad (12)$$

where $\mathcal{A}_{\text{lap}}$ enforces Laplacian spectral constraints, $\mathcal{A}_{\text{rel}}$ enforces relational smoothness and clock consistency, and $\mathcal{A}_{\text{ref}}$ constrains compatibility with a reference family. In implementation these constraints are audited through penalty functions $\{\Phi_k\}_{k \in \mathcal{K}}$ with calibrated thresholds $\{\theta_k\}$. These three constraint classes play different roles. Laplacian constraints prevent spectral objects that cannot arise from an admissible relational graph. Relational smoothness penalizes abrupt clock-slice variation that is not supported by the reference dynamics. Reference compatibility prevents the recovery from drifting toward a structurally valid but physically irrelevant spectral family. We keep these as separate penalties rather than collapsing them into a single score so that later policy decisions can identify whether a candidate failed because of structure, relational continuity, or reference mismatch.

In the experiments, Φ_{lap} is the sum over clock slices of structural Laplacian violations: symmetry error, positive-semidefinite violation, row-sum violation, positive off-diagonal violation, and diagonal-consistency error. Φ_{smooth} penalizes slice-to-slice spectral jumps by summing $\|\Lambda(\tau_{r+1}) - \Lambda(\tau_r)\|_2^2 / (\tau_{r+1} - \tau_r)^2$. Φ_{clock} compares spectral response between neighboring slices to the distinguishability of the corresponding clock states, and Φ_{ref} is the squared distance to the nearest reference trajectory. Thresholds are calibrated as $\mu + \kappa\sigma$ over the reference bank, with $\kappa = 1.5$ for the admissibility checks used in the encoded experiments. We use $\kappa = 1.5$ as a fixed operating tolerance for this baseline study; it is not optimized per regime.

For a candidate trajectory Γ_X , the normalized admissibility violation is

$$v_X := \max_{k \in \mathcal{K}} \max \left\{ 0, \frac{\Phi_k(\Gamma_X) - \theta_k}{\theta_k + \epsilon} \right\}. \quad (13)$$

Thus $v_X = 0$ means that all active constraints are within threshold, while positive values measure the worst normalized excursion beyond the admissible boundary.

Admissibility defines which spectral families are physically and structurally acceptable. It does not by itself provide a restoration direction. We therefore introduce a reference-anchor term that explicitly pulls the recovered trajectory toward the nearest compatible reference family. Given a reference bank $\mathcal{R}$, the anchor is

$$\Gamma_{\text{ref,anchor}} = \arg \min_{\Gamma' \in \mathcal{R}} d_{\mathcal{M}}(\Gamma', \Gamma_{\text{obs}})^2. \quad (14)$$

This anchor is the key distinction between feasibility and recovery. A projection onto $\mathcal{A}$ alone could return a trajectory that is merely allowed; it need not be the trajectory most compatible with the unperturbed relational behavior. The reference bank supplies a local restoration direction without requiring the method to know the clean trajectory of the test instance. In the encoded experiments, the bank contains three clean reference trajectories per code configuration: the corresponding logical reference state for the code configuration and two phase-rotated logical references at $\pm 20^\circ$. The bank is generated with the same code, clock grid, clock Hamiltonian, and graph functional as the test row, but with the noise schedule removed. For each observed trajectory, the anchor is selected using only $d_{\mathcal{M}}(\Gamma', \Gamma_{\text{obs}})$; no clean logical outcome, candidate fidelity, or row-level oracle information is used during anchor selection. The three-anchor bank is intentionally minimal. It tests whether a small auditable reference family can support recovery-side control, not whether a dense clean-state manifold has been approximated.

The recovery objective is

$$\begin{aligned} J_{\text{rec}}(\Gamma) = & \alpha d_{\mathcal{M}}(\Gamma, \Gamma_{\text{obs}})^2 \\ & + \beta d_{\mathcal{M}}(\Gamma, \Gamma_{\text{ref,anchor}})^2 \\ & + \lambda_1 \Phi_{\text{lap}}(\Gamma) + \lambda_2 \Phi_{\text{smooth}}(\Gamma) \\ & + \lambda_3 \Phi_{\text{clock}}(\Gamma). \end{aligned} \quad (15)$$

The first term keeps the recovered trajectory close to the observation, so the method does not discard the measured evidence. The second term pulls the solution toward the nearest compatible reference family, providing the actual restoration direction. The penalty terms then audit whether the resulting trajectory remains structurally and relationally admissible. This separation is deliberate: a candidate can be close to the observation but inadmissible, admissible but pointed toward the wrong reference family, or reference-compatible but too discontinuous across clock labels.

In the main experiments, we use the discrete admissible projection as the default recovery operator because it is stable, interpretable, and directly auditable. Continuous convex refinement is treated as a diagnostic extension and reported in the Supplementary Information. Further details

on admissibility penalties, reference-bank construction, anchor-weight calibration, and the two-stage recovery algorithm are provided in Supplementary Note S4.

3.4 Encoded-QEC validation protocol

The encoded-QEC setting is used as a controlled testbed for partial observability and unsafe recovery transitions. We do not claim that the proposed controller replaces optimized decoders for all QEC regimes. Instead, the testbed evaluates whether relational evidence can safely assist recovery when syndrome information is complete, missing, corrupted, ambiguous, or affected by measurement/reset errors. This framing is narrower than a decoder benchmark. The purpose of the encoded-QEC setting is to create a situation in which two plausible recovery candidates are available and the reliability of direct syndrome evidence can be systematically degraded. The scientific question is then not simply which candidate has the higher average fidelity, but when a policy should refuse a relational switch because the evidence supporting it has become unsafe.

Each simulation row produces two recovery candidates:

$$A := \text{syndrome-first decoded recovery}, \quad B := \text{relational admissible recovery}. \quad (16)$$

Candidate A is obtained by applying the code-specific syndrome recovery rule to the observed quantum trajectory. Candidate B is obtained by the admissible spectral recovery objective in Eq. (15). The policy layer then decides whether the relational candidate is sufficiently reliable to replace the syndrome-first candidate. Both candidates are evaluated against the same clean logical trajectory when computing outcomes, but the policy layer does not use that clean trajectory when making the decision. It sees only the candidate diagnostics, observed syndrome summary, admissibility reports, recovery objective, and uncertainty proxy. This separation is necessary because the intervention metrics later use the clean trajectory only as offline ground truth for analysis.

The experiments use three-qubit repetition codes in bit-flip and phase-flip configurations, with bit-flip, phase-flip, depolarizing, dephasing, and mixed-Pauli noise families. Each reported regime cell is averaged over seeds 11–20, giving ten random seeds per cell. The output-stem suffix `seed10` is a batch label for this ten-seed set. The encoding and syndrome-extraction circuits for both code configurations are shown in Supplementary Figs. S1 and S2. Syndrome observability is controlled by the observation ratio r_{obs} , syndrome-bit corruption probability p_{noise} , ambiguity level a_{amb} , measurement error probability p_{meas} , reset error probability p_{reset} , and measured syndrome information loss ℓ_{syn} . These quantities define the raw and clipped uncertainty proxies

$$u_{\text{raw}} = (1 - r_{\text{obs}}) + p_{\text{noise}} + a_{\text{amb}} + p_{\text{meas}} + p_{\text{reset}} + \ell_{\text{syn}}, \quad u_{\text{syn}} = \text{clip}_{[0,1]}(u_{\text{raw}}). \quad (17)$$

The uncertainty proxy is a deliberately simple additive stress index rather than a calibrated posterior uncertainty. The clipped value is used by the decision gate so that $u_{\text{max}} = 0.99$ implements a conservative saturation rule: saturated uncertainty blocks a proposed relational switch, while non-saturated uncertainty leaves the C_2 proposal eligible. The raw uncertainty is retained for diagnostics and bin-wise analysis. This distinction is important in the partial-plus-noisy and ambiguity-plus-measurement regimes, where several moderate degradations can combine into saturated decision uncertainty.

We compare four policies on the same candidates A and B . C_1 is a rule-based conservative hybrid that switches to B only when A is inadmissible and B is admissible, or when B is admissible, has lower recovery objective, and the observed syndrome is consistent. C_2 is the balanced proposal policy

$$S_X^{(2)} = \lambda_s s_X - \lambda_t d_X - \lambda_i (1 - a_X) + \lambda_o \frac{1}{1 + J_X}, \quad (18)$$

where a_X is the admissibility indicator, d_X is the trajectory distance to the observed trajectory, and s_X is the syndrome-support score. The current experiments use

$$(\lambda_s, \lambda_t, \lambda_i, \lambda_o) = (1.0, 1.0, 2.0, 0.5). \quad (19)$$

C_3 is a hard-admissibility ablation: if exactly one of A and B is admissible, it selects the admissible candidate; otherwise it falls back to a safety-weighted score. The initial C_3 policy was designed to test whether a globally strict safety-first region exists. Because this strict policy did not form an independent operating zone in the regime map, we use C_3^R as the revised risk-aware controller that preserves the safety motivation of C_3 while applying intervention only to proposed C_2 switches.

This policy sequence is intentionally cumulative. C_1 represents a conservative rule-based baseline. C_2 asks whether a balanced score can use relational evidence without excessive switching. C_3 tests whether a strict Boolean admissibility rule creates a distinct safety-first region. C_3^R is the revised controller used by the current study: it keeps C_2 as the proposal mechanism, but only allows its proposed switch to survive if additional safety and uncertainty gates pass.

The robust policy C_3^R treats C_2 as a proposal and applies a one-sided veto. Let $\Delta S = S_B^{(2)} - S_A^{(2)}$. The gate is

$$\begin{aligned} C_3^R = B &\iff [C_2 = B] \\ &\quad \wedge [\Delta S \geq \delta_S] \\ &\quad \wedge [v_A \geq \delta_A] \\ &\quad \wedge [B \text{ admissible}] \\ &\quad \wedge [v_B \leq \delta_B] \\ &\quad \wedge [u_{\text{syn}} \leq u_{\text{max}}]. \end{aligned} \tag{20}$$

Otherwise $C_3^R = A$. The main operating point is

$$\begin{aligned} \delta_S &= 0, & \delta_A &= 0, \\ \delta_B &= 0, & u_{\text{max}} &= 0.99. \end{aligned} \tag{21}$$

Thus C_3^R can only block C_2 switches; it never creates additional switches that C_2 rejected. The policy should therefore be interpreted as a downside-risk controller rather than a fourth independently tuned score rule. At the main operating point, the $v_A \geq \delta_A$ condition is nonrestrictive because $v_A \geq 0$ by definition. The active default gate therefore tests whether the C_2 -proposed relational candidate is score-supported, exactly admissible within the calibrated tolerances, and not exposed to saturated syndrome uncertainty. We keep the v_A term in Eq. (20) because it is part of the general gate family and can be activated in ablations. At this operating point, the uncertainty gate is the dominant veto mechanism in saturated-stress rows, while the exact-admissibility condition prevents accepting relational candidates that already violate the calibrated admissible set. Complete policy maps, decision reasons, and operating-zone summaries are provided in Supplementary Notes S5 and S6. The clean-regime role of C_3^R is correspondingly limited. If syndrome evidence is reliable and the uncertainty gate is open, the robust policy should often reduce to C_2 . Its value is tested primarily in regimes where syndrome evidence is partial, corrupted, or ambiguous enough that a proposed relational switch carries elevated downside risk.

3.5 Evaluation metrics and operating criteria

Since the controller is designed to suppress unsafe recovery transitions rather than maximize average fidelity alone, we evaluate both central performance and lower-tail risk. For each policy C , let

$$\Delta F_C = F_C - F_{\text{before}} \tag{22}$$

be the logical-fidelity gain relative to the noisy observed trajectory before recovery. Central performance is summarized by the mean gain $\overline{\Delta F_C}$, the logical-success rate $P(F_C \geq F_{\text{success}})$, and the non-worsening rate $P(\Delta F_C \geq 0)$. Unless otherwise stated, all reported comparisons use

$$\varepsilon_F = 0.01, \quad F_{\text{success}} = 0.99. \tag{23}$$

These thresholds are fixed before policy comparison and are not tuned per regime. Mean gain alone is not sufficient for this study. A policy could improve the mean by taking aggressive relational switches while still creating rare but severe regressions. Because the controller is intended to regulate unsafe transitions, we require metrics that expose the lower tail of the gain distribution and the quality of the blocks performed by C_3^R .

Downside risk is summarized by the fidelity false-safe rate

$$\text{FS}_F(C) = P[\Delta F_C < -\varepsilon_F], \tag{24}$$

the empirical lower quantile $q_{0.05}(C)$, and the lower conditional value at risk

$$\text{CVaR}_{0.05}(C) = \hat{\mathbb{E}}[\Delta F_C \mid \Delta F_C \leq q_{0.05}(C)]. \tag{25}$$

A successful risk controller should remain inactive when syndrome evidence is reliable, but should reduce false-safe and lower-tail failures under partial observability. We also use oracle regret as an offline diagnostic. For each row, define

$$\begin{aligned}\Delta F^* &= \max\{\Delta F_A, \Delta F_B\}, \\ \text{Reg}(C) &= \Delta F^* - \Delta F_C.\end{aligned}\tag{26}$$

This quantity does not define an implementable policy because it uses the row-level best candidate after the fact. It is nevertheless useful for measuring how far a policy falls from the best available candidate in the two-candidate decision problem.

For C_3^R , we additionally evaluate the quality of its interventions against the C_2 proposal. The block indicator is

$$B_R = \mathbf{1}[C_2 = B]\mathbf{1}[C_3^R = A].\tag{27}$$

A proposed C_2 switch is called harmful when

$$H_2 = \mathbf{1}[C_2 = B]\mathbf{1}[\Delta F_A > \Delta F_B + \varepsilon_F],\tag{28}$$

and beneficial when

$$G_2 = \mathbf{1}[C_2 = B]\mathbf{1}[\Delta F_B > \Delta F_A + \varepsilon_F].\tag{29}$$

We report the block rate $P(B_R = 1 \mid C_2 = B)$, harmful-block precision $P(H_2 = 1 \mid B_R = 1)$, harmful-switch recall $P(B_R = 1 \mid H_2 = 1)$, and beneficial-switch retention $P(B_R = 0 \mid G_2 = 1)$. The utility balance is

$$G_{\text{net}} = \sum_i B_R^{(i)} \max\{0, \Delta F_A^{(i)} - \Delta F_B^{(i)}\} - \sum_i B_R^{(i)} \max\{0, \Delta F_B^{(i)} - \Delta F_A^{(i)}\}.\tag{30}$$

Positive G_{net} means that, on the sampled grid, prevented harmful-switch losses outweighed missed beneficial-switch gains. It is a sample-level intervention diagnostic, not a theorem of dominance. We therefore also report normalized versions, including net effect per blocked switch and per C_2 -proposed switch, in the Supplementary Information. Full metric definitions are given in Supplementary Note S7, and uncertainty-threshold diagnostics are given in Supplementary Note S8. The final operating criterion is therefore qualitative as well as quantitative: C_3^R should leave reliable regimes nearly unchanged, block a substantial fraction of harmful C_2 switches under degraded syndrome evidence, and maintain a positive net intervention balance after accounting for beneficial switches that it mistakenly blocks.

4 Results

The results support an operating-zone interpretation rather than a universal performance claim. The relational controller is not beneficial in every regime. Its advantage appears specifically when syndrome observability is partial or jointly corrupted, where it suppresses unsafe recovery transitions and improves lower-tail fidelity risk. We therefore present the results by regime and intervention behavior, rather than ranking policies only by mean fidelity gain.

4.1 Operating-zone structure of the baseline policies

We first evaluated the clean-syndrome regime map to determine whether the baseline hybrid policies define distinct operating zones. The clean map contains 240 regime cells, each averaged over ten random seeds. As summarized in Table 1, C_1 is the preferred policy in most cells, while C_2 occupies a smaller but structured correction zone. The strict C_3 policy does not form an independent preferred region.

This baseline structure motivates the revised controller. The absence of a C_3 -dominant region does not mean that safety is irrelevant; it indicates that a globally hard admissibility veto is too coarse for this regime map. The redesign therefore shifts from a strict safety-first selector to C_3^R , an uncertainty-conditioned veto applied only to proposed C_2 switches. The C_2 -preferred cells are localized to phase-flip-code rows, with 16 dephasing, 10 phase-flip, 9 mixed-Pauli, 6 depolarizing, and 2 bit-flip cells. We use the clean map only to identify whether the original policy family contains distinct baseline operating zones. Degraded-regime behavior is evaluated separately through the C_2 versus C_3^R comparison below.

Table 1. Baseline policy operating-zone summary under reliable syndrome evidence. Preferred-policy counts are computed over 240 clean regime cells, with ten random seeds per cell.

Policy	Cells	Share
C_1	197	0.8208
C_2	43	0.1792
C_3	0	0.0000
C_3^R	0	0.0000

4.2 Regime-level effect of C_3^R

The regime-level comparison between C_2 and C_3^R is shown in Table 2. The most important pattern is selectivity: C_3^R remains inactive in the clean and noisy-only controls, but intervenes in partial and partial-plus-noisy syndrome regimes.

In the clean map, C_3^R leaves the C_2 behavior unchanged. The controller blocks none of the 1529 rows where C_2 selected the relational candidate. In the noisy-only map, it also remains inactive: C_2 and C_3^R have identical false-safe rate, lower-tail metrics, and logical success rate. This provides a negative control. The robust controller does not treat syndrome-bit corruption without observation loss as sufficient reason to impose a universal conservative bias. The noisy-only regime is not risk-free; its false-safe rate remains comparable to the partial regimes. However, because the observation ratio is not reduced and the uncertainty index does not saturate, C_3^R does not activate. This clarifies that the default controller is targeted to saturated partial-observability stress rather than to all forms of recovery risk.

Table 2. Regime-level comparison between C_2 and C_3^R . Each entry is reported as C_2/C_3^R . Lower false-safe rate, larger $q_{0.05}$, and larger CVaR indicate improved downside-risk behavior.

Regime	Mean	Success	Non-wors.	FS _F	$q_{0.05}$	CVaR _{0.05}
Clean	0.1273/0.1273	0.6358/0.6358	0.9554/0.9554	0.0238/0.0238	0.0000/0.0000	-0.0033/-0.0033
Noisy-only	0.1169/0.1169	0.4556/0.4556	0.8814/0.8814	0.0678/0.0678	-0.0104/-0.0104	-0.0207/-0.0207
Partial	0.1169/0.1189	0.4552/0.5976	0.8774/0.9306	0.0694/0.0389	-0.0152/-0.0006	-0.0208/-0.0147
Partial+noisy	0.1170/0.1212	0.4551/0.7241	0.8792/0.9755	0.0685/0.0134	-0.0104/0.0000	-0.0206/-0.0034
Ambig.+meas.	0.1600/0.1604	0.5208/0.5978	1.0000/1.0000	0.0000/0.0000	0.0192/0.0238	0.0078/0.0084

4.3 Downside-risk reduction under partial syndrome observability

The main benefit of C_3^R appears in the lower tail rather than in a large shift of the mean. In the partial-syndrome regime, the mean gain changes modestly from 0.1169 to 0.1189, while the false-safe fidelity rate decreases from 0.0694 under C_2 to 0.0389 under C_3^R . The lower 5% quantile moves from -0.0152 to -0.0006, while CVaR_{0.05} improves from -0.0208 to -0.0147. The logical-success rate increases from 0.4552 to 0.5976, and the non-worsening rate increases from 0.8774 to 0.9306.

The effect is stronger under partial-plus-noisy syndrome observability. There, C_3^R changes the mean gain from 0.1170 to 0.1212, but reduces the false-safe fidelity rate from 0.0685 to 0.0134 and shifts $q_{0.05}$ from -0.0104 to 0.0000. The lower-tail CVaR_{0.05} improves from -0.0206 to -0.0034. Thus the controller does not primarily act by maximizing average gain; it suppresses the negative tail of the recovery outcome distribution. This pattern is visualized in Fig. 2. The reduction should be interpreted together with the uncertainty-bin analysis in Supplementary Note S8: most blocks occur in saturated-stress rows, so the observed lower-tail improvement reflects a conservative veto of C_2 switches under high partial-observability stress. The relational component remains essential because the veto is applied only to C_2 -proposed switches into the relational candidate and is conditioned on the admissibility of that candidate.

The ambiguity-plus-measurement regime is informative for a different reason. Both C_2 and C_3^R have zero false-safe fidelity rate in this regime, so the binary false-safe metric alone would miss any distinction. The lower-tail metrics and logical-success rate nevertheless improve slightly under C_3^R , indicating that false-safe rate should be interpreted together with lower-tail and intervention diagnostics.

4.4 Intervention quality: harmful blocks, missed gains, and net effect

The previous subsection shows that C_3^R improves downside-risk metrics under partial observability. We next ask why. Because C_3^R can only veto proposed C_2 switches to B , the relevant quantities are conditional on $C_2 = B$.

In the partial-syndrome regime, C_2 selected B in 1094 rows and C_3^R blocked 476 of those switches, a conditional block rate of 0.4351. Among the blocked switches, the harmful-block precision was 0.8697: most blocked switches were cases where staying with A would have produced larger fidelity gain than switching to B . The prevented loss was 12.2670, while the missed beneficial gain was 6.6055, yielding a positive net intervention effect of 5.6616.

In the partial-plus-noisy regime, the intervention is stronger. C_3^R blocked 670 of 821 proposed C_2 switches to B , with harmful-block precision 0.8791 and harmful-switch recall 0.8475. The prevented loss was 17.4350 and the missed gain was 8.3139, giving a net intervention effect of 9.1212. This shows that the controller is not merely blocking more transitions. It blocks harmful transitions with high precision, and the prevented harmful loss exceeds the missed beneficial gain. The conditional intervention rates, aggregate balance, and normalized net effect across the active regimes are summarized in Fig. 3.

The cost of the controller is also visible. C_3^R blocks some beneficial switches: the beneficial-switch block rate is 0.3750 in the partial regime, 0.6739 in the partial-plus-noisy regime, and 0.4211 in the ambiguity-plus-measurement regime. The observed benefit therefore should not be described as free improvement. It is a risk-control tradeoff in which missed gains are accepted only when the prevented losses are larger on the sampled grid. The high beneficial-switch block rate in the partial-plus-noisy regime shows that C_3^R is not an efficiency-maximizing selector. Its operating point is intentionally conservative and should be evaluated by net downside-risk reduction, including normalized net effect, rather than by switch acceptance alone. This cost is substantial in the partial-plus-noisy regime, where more than half of beneficial C_2 switches are also vetoed. The present operating point should therefore be viewed as a safety-biased baseline rather than a tuned optimal controller.

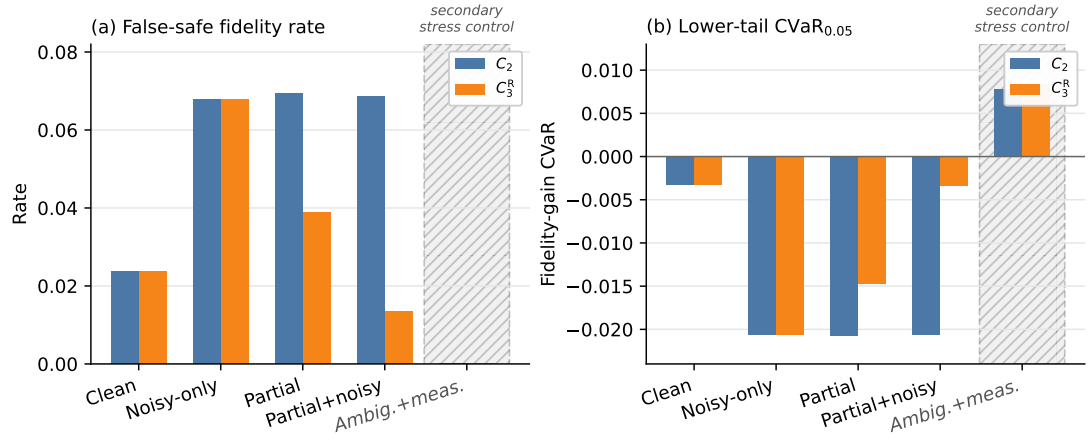


Figure 2. Downside-risk behavior of C_2 and C_3^R across syndrome regimes. Panel (a) shows the false-safe fidelity rate FS_F . Panel (b) shows the lower-tail fidelity-gain $CVaR_{0.05}$. The robust controller is inactive in clean and noisy-only controls, but reduces false-safe fidelity failures and improves lower-tail CVaR in partial and partial-plus-noisy regimes. The ambiguity-plus-measurement regime, hatched on the right, is included as a secondary stress control.

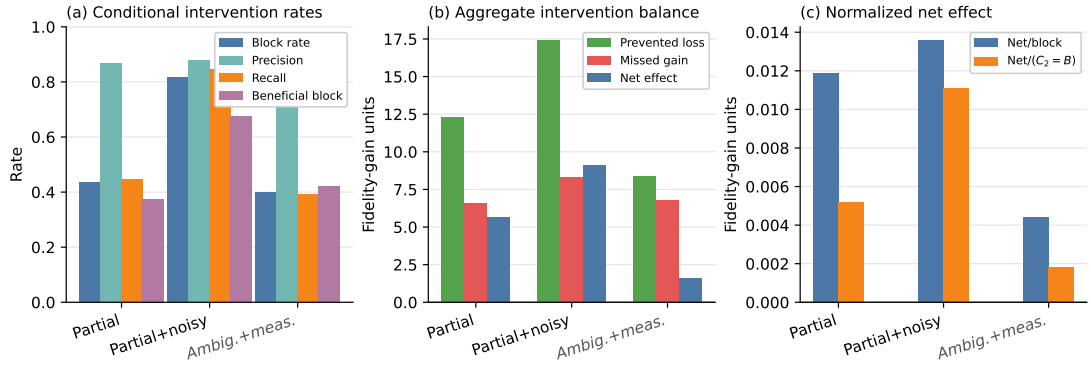


Figure 3. Intervention quality of C_3^R in regimes where it actively blocks C_2 switches. Panel (a) reports the conditional intervention rates: block rate (given $C_2 = B$), harmful-block precision, harmful-switch recall, and beneficial-switch block rate. Panel (b) shows the aggregate intervention balance, comparing prevented harmful-switch loss, missed beneficial-switch gain, and the resulting net effect, all in fidelity-gain units. Panel (c) normalizes the net effect per blocked switch and per C_2 -proposed switch to separate aggregate benefit from the number of vetoed proposals. The ambiguity-plus-measurement regime is hatched as a secondary stress control.

4.5 Negative controls and observed operating limits

The clean and noisy-only regimes serve as negative controls. In both cases, C_3^R is effectively inactive and does not materially change the C_2 outcomes. This is desirable: a risk-aware controller should not impose a universal conservative bias when the syndrome evidence is either reliable or only mildly corrupted without partial observability. The raw uncertainty mean is 0 in the clean map and 0.038 in the noisy-only map, far below the saturation condition that triggers uncertainty-based veto behavior.

The strongest observed operating zone is partial-plus-noisy syndrome observability, followed by partial syndrome observability. The ambiguity-plus-measurement regime provides weaker but consistent supporting evidence: C_3^R improves logical-success rate from 0.5208 to 0.5978 and yields positive net intervention gain, but the binary false-safe rate is already zero for both policies. These results should not be interpreted as a universal advantage of relational recovery over syndrome decoding. Rather, they identify a restricted but meaningful operating zone in which relational spectral evidence acts as a recovery-side safety signal under partial observability. Threshold diagnostics further show that setting $u_{\max} = 1.00$ removes the veto and recovers C_2 behavior, whereas stricter $u_{\max} = 0.75$ increases blocking and can reduce FS_F further at the cost of more beneficial-switch loss. In some regimes this becomes overly conservative. Thus the default point is not claimed to be optimal; it is a conservative operating point that exposes the downside-risk tradeoff.

5 Discussion

5.1 Operating-zone interpretation of relational recovery

The results support an operating-zone interpretation rather than a universal dominance claim for relational recovery. Relational spectral evidence is useful in the tested setting when syndrome information is sufficiently incomplete to make some C_2 switches unsafe, but not so uninformative that the relational trajectory ceases to provide a meaningful consistency signal. This explains why C_3^R is inactive in clean and noisy-only controls, yet changes the lower tail in partial and partial-plus-noisy regimes. The roles of the two components should be separated. The relational trajectory supplies candidate B , admissibility status, and trajectory diagnostics. The uncertainty gate determines when a C_2 -proposed use of that candidate is too risky to accept.

The negative strict- C_3 result is part of this interpretation. A hard admissibility rule does not create an independent preferred zone under reliable syndrome evidence. Safety therefore cannot be reduced to selecting the uniquely admissible candidate. The revised controller uses a narrower principle: keep the balanced C_2 proposal mechanism, but veto proposed relational switches when the proposal sits in a high-risk part of the policy evidence space. The main claim is therefore not that C_3^R is a better decoder, but that it is a risk-aware controller for candidate selection under partial observability. Read as part of the baseline contribution, this negative ablation also serves a constructive purpose: it establishes that relational spectral evidence cannot be collapsed into a simpler Boolean admissibility check, which justifies the more elaborate one-sided veto structure that the present work and its planned follow-ups build on.

5.2 Partial observability and decision geometry

The relevant geometry is the geometry of the decision variables seen by the policy. Each row supplies a score difference, admissibility status, normalized constraint violations, and an additive syndrome-stress index. In clean regimes, direct syndrome support keeps the C_2 proposal away from the veto boundary. In noisy-only regimes, bit-level syndrome noise changes some scores but does not produce the saturated uncertainty pattern that activates the robust gate. Partial observability is different: the score can favor the relational candidate while the same row has weak syndrome support and elevated uncertainty.

This is the failure mode targeted by C_3^R . The controller does not penalize all relational switches, and it does not treat all noise as equivalent. It acts only after C_2 proposes B , and then requires score support, admissible B , non-saturated uncertainty, and the configured violation gate. At the default setting $\delta_A = 0$, the active requirements reduce primarily to a score-supported C_2 proposal, exact admissibility of B , and the uncertainty saturation guard. This makes the downside-risk policy mathematically one-sided: it can remove candidate- B switches from the C_2 decision set, but it cannot create new relational switches.

The intervention metrics show the cost of this one-sided geometry. Some beneficial switches are blocked, especially in the active degraded regimes. The reason the policy remains useful in these experiments is that the prevented harmful-switch loss exceeds the missed beneficial gain, and this effect appears most clearly in the lower-tail metrics. The conclusion should remain conservative: the present evidence supports a downside-risk reduction mechanism on the sampled grids, not a proof that the same fixed gate is optimal across codes or noise models, nor that saturated uncertainty is the only useful trigger.

5.3 Recovery control rather than decoder replacement

The QEC role of the method is a recovery-control layer above candidate recovery mechanisms. Candidate A is the syndrome-first recovery, candidate B is the relational admissible recovery, and the policy selects between them using diagnostics that are available before the logical outcome is evaluated. This is a different task from constructing an optimized syndrome decoder.

This distinction sets the correct benchmark for the present study. Mean fidelity alone is insufficient because the controller is designed to alter the lower tail of candidate selection, not to maximize every row independently. The relevant questions are whether reliable regimes remain nearly unchanged, whether harmful C_2 switches are selectively vetoed under degraded syndrome evidence, and whether the missed beneficial switches are smaller than the prevented losses. The current results answer these questions for the controlled two-candidate setting. Comparisons with optimized decoders and multi-candidate recovery sets remain future work. A fair next benchmark should compare not only final logical fidelity, but also false-safe rate, lower-tail gain, and intervention regret under matched partial-observability protocols.

The two-candidate restriction is itself a deliberate baseline choice rather than a structural limitation of the framework. The decision geometry of a one-sided veto is most clearly

characterized when only one alternative to the syndrome-first candidate is available: the conditional metrics in Section 4.4, including block rate, harmful-block precision, and beneficial-switch retention, are well defined exactly because the policy maps a binary choice. Extending to a multi-candidate recovery set replaces these binary diagnostics with class-conditional analogues but does not alter the underlying veto principle, which acts on the C_2 -proposed switch regardless of how many alternatives the proposal layer considers. We therefore treat A/B as the smallest setting in which the intervention quality of C_3^R can be cleanly read off, and leave multi-candidate extension to subsequent work.

5.4 Clock-conditioned interpretation and spectral geometry

The background-independent component enters as a data-object construction. The global constraint motivates clock-conditioned system slices, and those slices are converted into relational graph spectra. The policy never uses an external time-indexed trajectory; it uses an ordered family of internally clocked spectral objects and calibrated admissibility penalties. The conceptual significance is that the recovery evidence is indexed by internal relational labels rather than by an externally supplied trajectory variable, which is what allows admissibility, smoothness, and clock-consistency penalties to be defined in a single relational language.

Operationally, the Page–Wootters-type construction is used here as a finite data-object construction rather than as a foundations-level theory of quantum time. The metric trajectory representation denotes a constrained finite trajectory space equipped with a metric, a reference bank, and admissibility tests. That much is sufficient to produce a recovery-side signal: a proposed candidate can be checked for compatibility with clock-conditioned relational structure, and the resulting decision is auditable row by row. We retain the relational framing not because it is necessary for the present numerical experiments in isolation, but because it gives a structured component interface — clock register, conditional slice, graph functional, spectral trajectory, reference bank, admissibility penalty — along which subsequent work can substitute richer relational representations or learned structural maps without disturbing the surrounding decision geometry. In this sense the relational layer provides a structured interface for planned follow-up studies rather than a self-contained claim about quantum time.

5.5 Limitations and outlook

We organize remaining limitations as three follow-up tracks, each of which corresponds to a separate planned study. The first track is QEC benchmarking: larger codes, broader channel families, optimized syndrome decoders, and multi-candidate recovery sets are needed before making stronger decoder-level claims [40, 41, 42]. The present work establishes a controlled baseline for two-candidate selection, not a competitive decoder benchmark, and the result also depends on the two-candidate formulation A/B ; a larger candidate set may change the intervention tradeoff. Whether the same operating-zone and downside-risk geometry persist when the relational signal is applied to nontrivial stabilizer codes such as the $[[5, 1, 3]]$ perfect code, the Steane code, and the Shor nine-qubit code is the subject of forthcoming work.

The second track is relational representation. The graph functional is fixed in the reported experiments, with mutual information used for the bit-flip code and absolute two-qubit coherence used for the phase-flip code. Alternative functionals, larger reference banks, multipartite relational witnesses, and learned relational summaries may change the sensitivity of the spectral trajectory and should be tested explicitly [43, 44, 45].

The third track is uncertainty and control. The current uncertainty proxy is an additive stress index, not a calibrated posterior. The default $u_{\max} = 0.99$ gate is best interpreted as a saturation guard, and the normalized intervention metrics show that the gate trades missed beneficial switches against prevented harmful switches. Future work should calibrate this uncertainty more directly, learn adaptive thresholds, and test whether the same downside-risk geometry persists beyond the sampled regimes and under pulse- or control-level optimization constraints [46, 47]. We also stress-tested the admissibility threshold itself by varying $\kappa \in \{1.0, 1.5, 2.0\}$ across a balanced grid of 135 recovery cases (see Supplementary Note S8): admissibility coverage saturates at $\kappa = 1.5$ and remains stable up to $\kappa = 2.0$, while $\kappa = 1.0$ leaves most recovered trajectories outside the calibrated admissible set. The default $\kappa = 1.5$ should therefore be read as the smallest tolerance at which admissibility coverage saturates on the sampled configurations, not as a tuned optimum, and a joint $(\kappa, u_{\max})$ calibration on larger codes is part of the planned follow-up program. The longer-term direction joining the second and third tracks is to explore whether the relational spectral trajectory space, used here as an auditable diagnostic object, may support learned structural representations of admissible recovery geometry.

Taken together, the evidence supports a cautious claim: clock-conditioned relational spectra can serve as an operational safety signal for recovery under partial quantum observability. The present study does not establish a universal decoder or a complete background-independent dynamics. It provides a reproducible finite-qubit setting in which relational spectral admissibility, partial observability, and downside-risk recovery control can be studied together.

6 Conclusion

We introduced a finite, auditable framework for clock-conditioned relational spectral recovery in partially observable qubit networks, and evaluated it as a risk-aware controller for candidate selection in encoded quantum-error-correction settings. The methodological core is the mapping of clock-conditioned system slices to weighted relational graphs and the use of the resulting Laplacian spectral trajectory as recovery-side evidence. Within this framework, recovery is posed as constrained restoration toward an anchored reference family, with admissibility penalties that the policy layer can verify row by row. The risk-aware controller C_3^R is constructed as a one-sided veto on the balanced proposal policy C_2 , so its intervention can be analyzed on the same two-candidate decision problem without changing how the candidates are constructed.

The empirical evaluation supports a restricted but reproducible claim. C_3^R leaves clean and noisy-only regimes effectively unchanged, yet reduces false-safe fidelity rates and lower-tail risk under partial and partial-plus-noisy syndrome observability, where the prevented harmful-switch loss exceeds the missed beneficial gain on the sampled grids. The results are therefore consistent with an operating-zone interpretation: relational spectral structure acts as a recovery-side safety signal precisely when syndrome evidence is incomplete enough to make balanced score-based switches unsafe, but not so uninformative that the relational trajectory loses its consistency value. The negative result for the strict admissibility ablation C_3 further indicates that safety cannot be reduced to a Boolean choice between admissible candidates once inadmissibility has already been priced into a balanced score.

The contribution should be read narrowly. We do not claim a universal decoder, an optimized risk controller, or a complete background-independent quantum dynamics. The Page-Wootters-type construction is used only as a data-object construction for finite encoded simulations, the uncertainty proxy is an additive stress index rather than a calibrated posterior, and the default operating point is deliberately conservative rather than tuned. What the framework provides is a controlled, auditable setting in which relational spectral admissibility, partial observability, and downside-risk recovery control can be studied jointly, with explicit operating-zone, intervention-quality, and threshold diagnostics. This positions relational spectral evidence as a complementary safety layer above existing recovery and mitigation pipelines for encoded qubit networks, rather than as a competitor to optimized decoders. Extending the protocol to larger codes, optimized syndrome decoders, multi-candidate recovery sets, alternative graph functionals, and calibrated uncertainty models is a natural next step, and the present design is intentionally structured to support such extensions without changing the underlying decision geometry. We accordingly intend the present study as the first of a planned sequence: a follow-up will test whether the same admissibility and risk-control geometry generalizes to nontrivial stabilizer codes, while a longer-term direction explores whether the same trajectory space may support learned structural representations of admissible recovery geometry. The component interfaces defined here are chosen with both extensions in mind.

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Competing interests

The authors declare no competing interests.

Data availability

The source code, simulation scripts, and figure-generation assets used in this study are publicly available at <https://github.com/Jeong-Jaeyun/Risk-Aware-Relational-Spectral-Recovery-under-Partial-Quantum-Observability.git> and archived on Zenodo (DOI: 10.5281/zenodo.19471034). Additional intermediate simulation outputs are available from the corresponding author upon reasonable request.

Author contributions

J.Y.J. conceived the study, implemented the simulations, performed the data analysis, and drafted the manuscript. H.Y.J. supervised the research, provided critical intellectual input, and revised the manuscript. All authors have read and approved the final version of the manuscript.

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